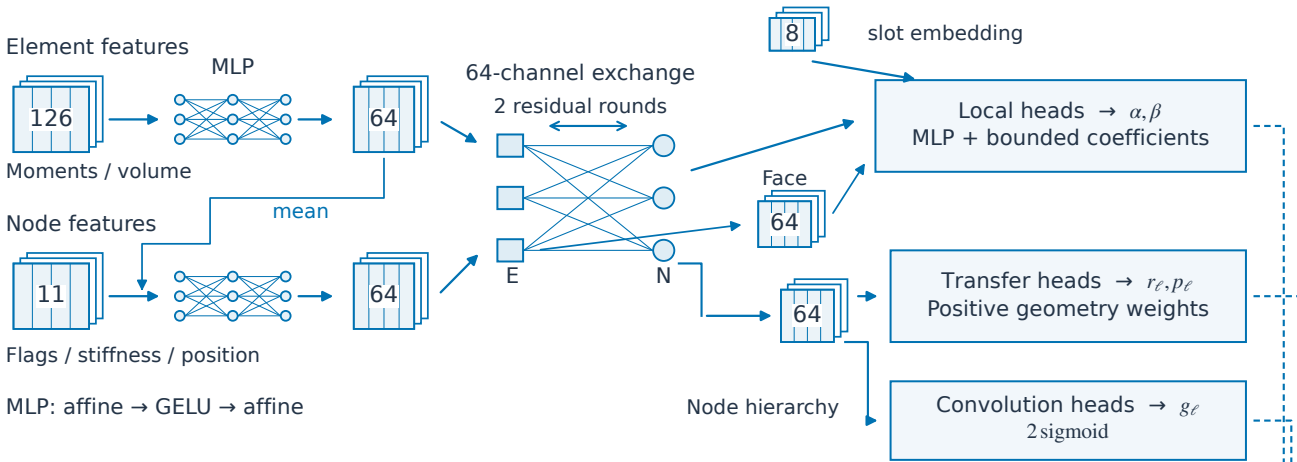
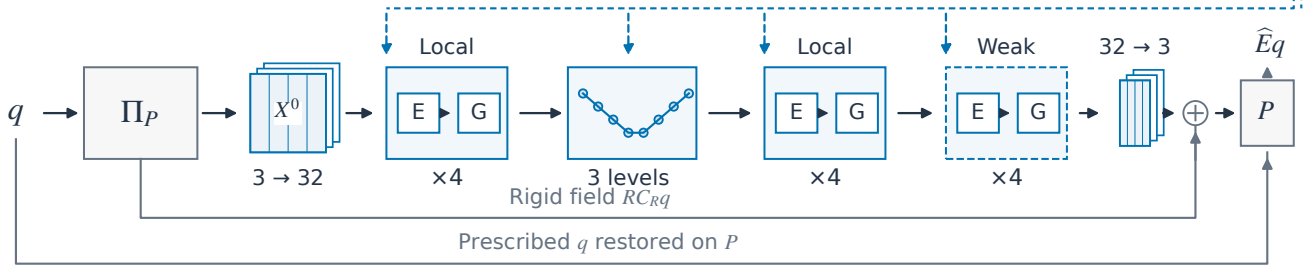


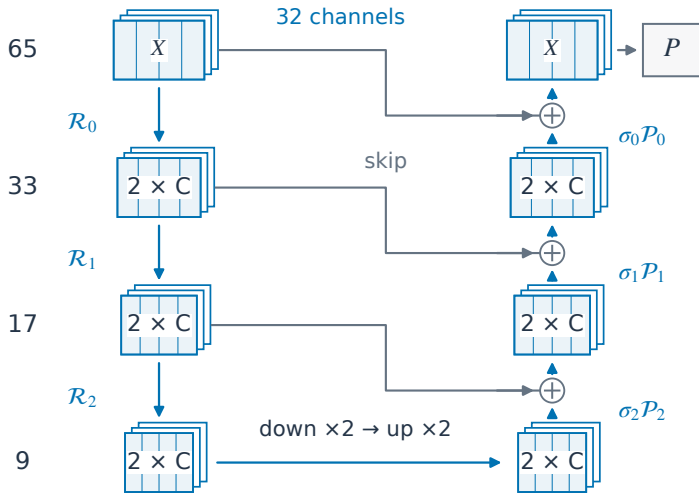
(a) Geometry hypernetwork



(b) Displacement branch: linear in q for fixed geometry



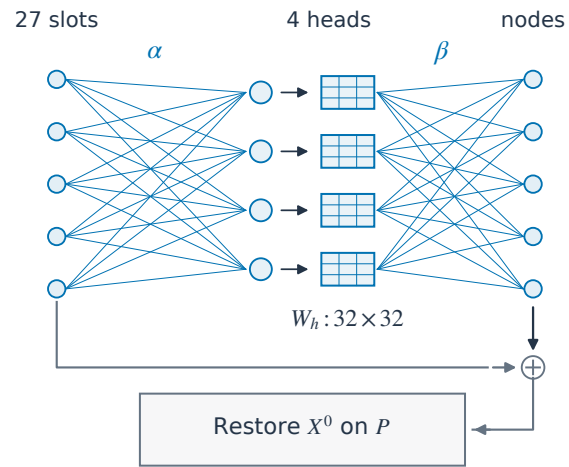
(c) Multiscale displacement propagation



R : restriction P : prolongation

$C: X \leftarrow X + g \odot \text{Conv}_{3 \times 3 \times 3}(X)$

(d) Local interaction



E: 27 Q2 nodes

G: 18 owner + 9 neighbour nodes

Training: update θ through both branches.

Inference: fix θ ; cache geometry coefficients.